

Calibration procedures for object locating sensors in flexible robotized machining

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Abstract— We present calibration procedures for localization sensor systems for flexible robotic machining. For the needs of flexible production with varying parts and locations, object detection and localizing sensor systems carry out rough global localisation with a laser profiler or a multi-camera system. For needs of machining, more accurate and local object and surface localisation is carried out with a wrist mounted high resolution laser profiler. The calibration procedures for these sensors were implemented and tested with an industrial robot. The novelty of our work is in relying on simple calibration objects, enabling easy and flexible calibration procedures.

Keywords— robot manipulators, calibration, sensors

I. INTRODUCTION

Interest for use of robots or manipulators for machining hard materials tasks has increased recently [1]. Major advantages of robots in machining are higher flexibility, lower investment costs and larger working space. Production flexibility in robotic machining system comes with sensors for adapting to varying shapes and locations of machined objects. The sensory systems typically include laser profilers/scanners, imaging sensors and force/torques sensor. In order to develop cost effective robotised machining solutions, the costs of required sensors are of great importance as well as automatic and user friendly calibration and commissioning procedures.

Typical sensor based operations in robotic machining are object detection and localisation, on-line path planning and compliant motion control. Here we focus on sensors which can be located in a fixed location in the robot cell, or in the robot wrist. The sensor can operate globally, i.e. detect, recognize and locate target objects in the whole robot work space, or locally, i.e., locate target features or target objects in local space with short distance. For global work space localization the range is in metres and accuracy is in the level of several millimetres or centimetres. In local work space the range is in centimetres or tens of centimetres, and accuracy in a sub-millimetre level. The coordinate frames of the sensors need to be well calibrated to the robot to minimize the systematic errors in the robot path accuracy.

In our approach global localisation can be done in two ways. Automated object detection and localizing sensor system carries out the global localisation with a laser profiler or alternatively we use also interactive object localization [2] where an operator shows characteristics points directly from real object surfaces. Then local localisation with a wrist

mounted high resolution laser profiler finds the details of the target object.

In this paper we present a set of calibration procedures which enable easy and flexible calibration of localization sensor systems for robotic machining. The procedures are partly based on earlier work from literature, and here we take a further look on their implementation and practical tests. The principles for a hand mounted sensor and interactive multi-camera system are found from [3] and [4]. The calibration procedure for the fixed locating sensor system is new and follows the principles for an eye-in-hand sensor calibration given in [5]. Chapter II presents the calibration procedures with literature references, chapter III test results and discussions and conclusions are given in chapter IV.

II. CALIBRATION PROCEDURES

A. Calibration of a Eye-in-Hand 2D laser scanner

Calibration of the location of a 2D laser scanner (profiler) attached to the robot wrist— transformation “x” in Fig. 1 - means solving the equation of the form

$$Ax = xB \quad (1)$$

in which A is the transformation from i:th to j:th gripper pose in the robot base coordinate system and B is the corresponding transformation of the sensor in the calibration block coordinate system. Generally this equation does not have a closed form solution, but there are solutions given in the literature separating the rotation and translation part of the transformation and solving the rotational part first, see for example [3] and [7]. Here we apply the seminal algorithm by Tsai & Lenz [3] where the observables are poses of the mounting flange in the robot coordinate system and the scanner poses in a stationary calibration object coordinate system. To carry out the hand/eye calibration a stationary calibration block is scanned in five unique poses. The pose of the scanner with respect to the calibration block is estimated by fitting a planar CAD model of the calibration block to the measured profiles of the laser scanner. Several pose pairs are attained, represented in the robot base coordinates and in the calibration object coordinate frame. By observing the changes between pairs of poses we get local pose-to-pose motions, which physically share the same rotation axis. The rotation part of the estimated transformation is presented by a rotation axis, scaled with the magnitude of the rotation, which can be presented in linear form and solved in closed form. After the rotation has been solved, the translational part can be solved trivially.

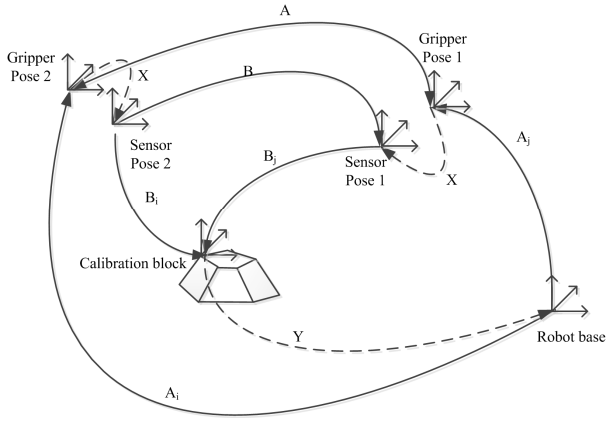


Fig. 1. Calibration principle for the Eye-in-Hand sensor

The procedure for computing the rotation matrix from scanner to the robot flange goes as follows [3]:

Step 1: Solve principal vector defining rotation from scanner coordinate system to the robot flange coordinate system P_{cg} from a system of linear equations.

$$Skew(P_{gij} + P_{cij})P_{cg}' = (P_{cij} - P_{gij}) \quad (2)$$

in which P_{cij} is the principal vector defining rotation from i :th to j :th scanner pose and P_{gij} define the respective robot flange rotations.

The principal rotation vector P_r is defined as

$$P_r = 2\sin\frac{\theta}{2}[n1 \ n2 \ n3]^T \quad (3)$$

in which θ is the angle of rotation in the range $0 \leq \theta \leq \pi$ and $n1, n2, n3$ are the components of the unit vector defining the rotation axis.

Skew-symmetric matrix is generated by a 3D vector

$V = [vx \ vy \ vz]^T$ and is as follows

$$Skew(V) = \begin{bmatrix} 0 & -vz & vy \\ vz & 0 & -vx \\ -vy & vx & 0 \end{bmatrix} \quad (4)$$

Step 2: Compute principal vector P_{cg} defining rotation from scanner to the robot flange

$$P_{cg} = \frac{2P_{cg}'}{\sqrt{1 + |P_{cg}'|^2}} \quad (5)$$

and the corresponding rotation matrix R_{cg} is calculated as follows

$$R_{cg} = \left(1 - \frac{|P_{cg}|^2}{2}\right) I + \frac{1}{2}(P_{cg} \cdot P_{cg}^T + \alpha \cdot skew(P_{cg})) \quad (6)$$

in which I is a unit matrix of size $[3 \times 3]$ and

$$\alpha = \sqrt{4 - |P_r|^2} \quad (7)$$

Step 3: Solve translation vector T_{cg} from scanner to the robot flange from a system of linear equations

$$(R_{gij} - I)T_{cg} = R_{cg}T_{cij} - T_{gij} \quad (8)$$

in which T_{cij} is a translation vector from i :th scanner pose to j :th scanner pose and T_{gij} is a translation vector from i :th flange pose to j :th flange pose. R_{cg} is a rotation matrix calculated in step 2.

At least two pose pairs i.e. three scanning stations are needed and having more improves the accuracy of the result. Using this fact the rotational parameters can be represented in linear form and solved. More detailed description of the algorithm can be found in [3].

B. Calibration of a fixed 3D laser scanner

In our 3D laser scanner a 2D laser profiler is attached to a straight rail. 3D points from target object's surfaces are measured by moving the scanner along the rail and scanning at the same time. The sensor system must express the scanned target object surface points in robot's base coordinate frame. Automatic calibration of the 3D laser scanner can be carried out with a special but very simple calibration object: a planar calibration target object, which the robot moves in front of the laser scanner, basically in arbitrary number of arbitrary locations. 3D points are measured from the calibration object surface in these different locations. The calibration measurement data is composed of the set of these 3D points in the coordinate frame of the laser scanner, and corresponding set of robot locations in robot base coordinate frame. In Fig. 2 the coordinate systems related to the calibration are illustrated.

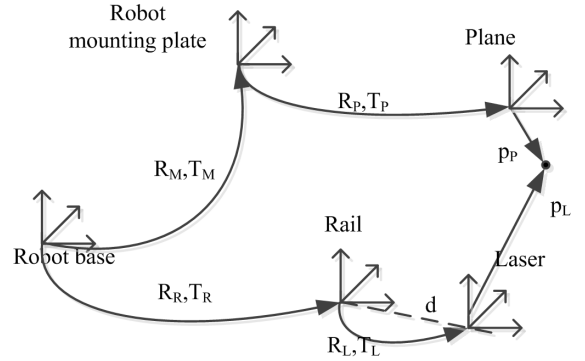


Fig. 2. Coordinate frames and transformations with rotation matrixes (R) and translation vectors (T).

Mathematically the problem is an extended pose estimation problem: to determine the coordinate transformations between the laser scanner and the rail and between the robot and the rail. Our solution goes back to the ideas first presented by Lowe [8], where 2D point measurements from known target object image features are fitted to an object model with iterative estimation. Since Lowe numerous successful examples have shown with varying types of measurements the feasibility of the approach for acquiring an optimal pose estimate for the object location - provided that correspondence with the measured points and object model features is solved reliably. These include 1D measurements synchronized to robot motions to determine grasping errors [6], calibration of multiple cameras [4] and

calibration of a range sensor in a robot wrist to the robot coordinate system [5].

The novelty of our approach is in extending conventional pose estimation with one translation and two rotation parameters, i.e. the location of the calibration plate in the robot flange coordinate frame. The calibration plate has origin its' plane and z-axis is the corresponding plane normal. The accurate location of the origin in the plate is not relevant.

The parameters to be estimated are pose parameters of the rail including two rotations

$$\boldsymbol{\varphi}_R = \begin{bmatrix} \varphi_{Rx} \\ \varphi_{Rz} \end{bmatrix} \quad (9)$$

and three translations parameters

$$\boldsymbol{T}_R = \begin{bmatrix} t_{Rx} \\ t_{Ry} \\ t_{Rz} \end{bmatrix} \quad (10)$$

Three rotational pose parameters of the laser scanner

$$\boldsymbol{\varphi}_L = \begin{bmatrix} \varphi_{Lx} \\ \varphi_{Ly} \\ \varphi_{Lz} \end{bmatrix} \quad (11)$$

and pose parameters of the calibration plate including two rotations

$$\boldsymbol{\varphi}_P = \begin{bmatrix} \varphi_{Px} \\ \varphi_{Py} \end{bmatrix} \quad (12)$$

and one translation parameters

$$\boldsymbol{T}_P = \begin{bmatrix} 0 \\ 0 \\ z_p \end{bmatrix} \quad (13)$$

With linearization the eleven parameters can be estimated iteratively. At least 5 profile lines are needed; more lines improve accuracy. The correction values of the pose parameters are collected to a column matrix $\boldsymbol{c}$:

$$\boldsymbol{c} = [\Delta\varphi_{Rx} \ \Delta\varphi_{Rz} \ \Delta t_{Rx} \ \Delta t_{Ry} \ \Delta t_{Rz} \ \Delta\varphi_{Lx} \ \Delta\varphi_{Ly} \ \Delta\varphi_{Lz} \ \dots \ \Delta\varphi_{Px} \ \Delta\varphi_{Py} \ \Delta t_{Pz}]^T \quad (14)$$

Fig. 2 shows the coordinate frames with the following coordinate transformations considered:

The laser scan points in its xz-plane. Hence, measured points $\boldsymbol{p}_L$ have zero y-coordinates.

- The frame of the rail is set so that the origin of the laser scanner's frame moves along its positive y-axis when laser is moved along the rail. The movement distance is obtained as measurement d . The translation vector $\boldsymbol{T}_L$ from rail's frame to laser's frame is therefore known and needs not be calibrated. The rotation matrix $\boldsymbol{R}_L$ needs to be fully calibrated (three rotations).
- The position of the rail frame's origin in robot's base frame, i.e. $\boldsymbol{T}_R$, must be fully calibrated (three coordinates). The orientation of rail's frame, i.e. $\boldsymbol{R}_R$, must be calibrated

only concerning the direction of the y-axis. Hence, only rotations around x- and z-axes are corrected in calibration.

- The position and orientation of the robot's mounting plate's frame in robot's base frame, i.e. $\boldsymbol{T}_M$ and $\boldsymbol{R}_M$, are known.
- For calibration measurements the robot is carrying a plane surface. The coordinate frame of this plane is chosen so that its xy-plane coincides with the plane surface and z-axis points away from the surface. The position of the plane in the robot mounting plate frame can move freely along its xy -plane. Hence, its position $\boldsymbol{T}_P$ is corrected in calibration only in its z-direction. The calibration plane can also rotate freely around its z-axis and its orientation $\boldsymbol{R}_P$ is corrected only by rotating around x- and y-axes.

Below equations are shown for transforming the measured point from laser's frame to calibration plane's frame.

- Measured point on plane in laser's frame:

$$\boldsymbol{p}_L = \begin{bmatrix} x_L \\ 0 \\ z_L \end{bmatrix} \quad (15)$$

- Point transformed to rail's frame:

$$\boldsymbol{p}_R = \boldsymbol{R}_L \boldsymbol{p}_L + \boldsymbol{T}_L \quad (16)$$

$$\boldsymbol{T}_L = \begin{bmatrix} 0 \\ -d \\ 0 \end{bmatrix} \quad (17)$$

- Point transformed to robot's base frame:

$$\boldsymbol{p}_0 = \boldsymbol{R}_R \boldsymbol{p}_R + \boldsymbol{T}_R \quad (18)$$

- Point transformed to robot's mounting plate's frame:

$$\boldsymbol{p}_M = \boldsymbol{R}_M^T (\boldsymbol{p}_0 - \boldsymbol{T}_M) \quad (19)$$

- Point transformed to plane's frame:

$$\boldsymbol{p}_P = \boldsymbol{R}_P^T (\boldsymbol{p}_M - \boldsymbol{T}_P) \quad (20)$$

In the calibration procedure the calibration plane is moved to several different positions and orientations under the rail (i.e. different robot angles) and points are scanned from its surface with several positions of the laser on the rail. Initial estimates for all relations between coordinate frames are set and all measured points are transformed to calibration plate frame. Because of errors in frame relation estimates, these points in plane frame are not exactly in plane frame's xy-plane, they have non-zero z-coordinates. These z-coordinate values are used as error quantities e which are collected to a column matrix $\boldsymbol{E}$ and the corresponding derivatives $\partial e / \partial \boldsymbol{c}$ are collected row by row to Jacobian matrix $\boldsymbol{J}$.

The estimates for corrections in $\boldsymbol{c}$, which minimize the square sum of error quantities, i.e. $\boldsymbol{E}^T \boldsymbol{E}$, can be computed as:

$$\boldsymbol{c} = -(\boldsymbol{J}^T \boldsymbol{J})^{-1} \boldsymbol{J}^T \boldsymbol{E} \quad (21)$$

The derivative of the error quantity e with respect to all correction values $\boldsymbol{c}$ is a row consisting of the last rows of derivatives of point $\boldsymbol{p}_P$:

$$\frac{\partial e}{\partial \boldsymbol{c}} = [0 \ 0 \ 1] \begin{bmatrix} \frac{\partial \boldsymbol{p}_P}{\partial \varphi_R} & \frac{\partial \boldsymbol{p}_P}{\partial \Delta \boldsymbol{T}_R} & \frac{\partial \boldsymbol{p}_P}{\partial \varphi_L} & \frac{\partial \boldsymbol{p}_P}{\partial \varphi_P} & \frac{\partial \boldsymbol{p}_P}{\partial \Delta z_P} \end{bmatrix} \quad (22)$$

$$\frac{\partial \mathbf{p}_P}{\partial \boldsymbol{\varphi}_R} = \mathbf{R}_P^T \mathbf{R}_M^T \mathbf{R}_R \begin{bmatrix} 0 & -y_R \\ -z_R & x_R \\ y_R & 0 \end{bmatrix} \quad (23)$$

$$\frac{\partial \mathbf{p}_P}{\partial \Delta \mathbf{T}_R} = \mathbf{R}_P^T \mathbf{R}_M^T \mathbf{R}_R \quad (24)$$

$$\frac{\partial \mathbf{p}_P}{\partial \boldsymbol{\varphi}_L} = \mathbf{R}_P^T \mathbf{R}_M^T \mathbf{R}_R \mathbf{R}_L \begin{bmatrix} 0 & z_L & -y_L \\ -z_L & 0 & x_L \\ y_L & -x_L & 0 \end{bmatrix} \quad (25)$$

$$\frac{\partial \mathbf{p}_P}{\partial \boldsymbol{\varphi}_P} = \begin{bmatrix} 0 & -z_P \\ z_P & 0 \\ -y_P & x_P \end{bmatrix} \quad (26)$$

$$\frac{\partial \mathbf{p}_P}{\partial \Delta z_P} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \quad (27)$$

Below the corrections to position and orientation relations are shown, which are computed in calibration. Also the derivatives of a measured point transferred to plane's frame with respect to correction values are shown. These derivatives are computed at zero values for corrections.

- Correcting the rail's position and orientation in robot's base frame:

$$\mathbf{R}_R \leftarrow \mathbf{R}_R \Delta \mathbf{R}_R \quad (28)$$

$$\mathbf{T}_R \leftarrow \mathbf{R}_R \Delta \mathbf{T}_R + \mathbf{T}_R \quad (29)$$

$$\Delta \mathbf{R}_R = \mathbf{Rot}_x(\Delta \varphi_{Rx}) \mathbf{Rot}_z(\Delta \varphi_{Rz}) \quad (30)$$

$$\Delta \mathbf{T}_R = \begin{bmatrix} \Delta t_{Rx} \\ \Delta t_{Ry} \\ \Delta t_{Rz} \end{bmatrix} \quad (31)$$

- Correcting the laser's orientation in rail's frame:

$$\mathbf{R}_L \leftarrow \mathbf{R}_L \Delta \mathbf{R}_L \quad (32)$$

$$\Delta \mathbf{R}_L = \mathbf{Rot}_x(\Delta \varphi_{Lx}) \mathbf{Rot}_y(\Delta \varphi_{Ly}) \mathbf{Rot}_z(\Delta \varphi_{Lz}) \quad (33)$$

- Correcting the plane's position and orientation in robot's mounting plate's frame (tool correction):

$$\mathbf{R}_P \leftarrow \mathbf{R}_P \Delta \mathbf{R}_P \quad (34)$$

$$\mathbf{T}_P \leftarrow \mathbf{R}_P \Delta \mathbf{T}_P + \mathbf{T}_P \quad (35)$$

$$\Delta \mathbf{R}_P = \mathbf{Rot}_x(\Delta \varphi_{Px}) \mathbf{Rot}_y(\Delta \varphi_{Py}) \quad (36)$$

$$\Delta \mathbf{T}_P = \begin{bmatrix} 0 \\ 0 \\ \Delta z_P \end{bmatrix} \quad (37)$$

The corrections to $\mathbf{R}_R$, $\mathbf{T}_R$, $\mathbf{R}_L$, $\mathbf{R}_P$ and $\mathbf{T}_P$ are realized and the whole procedure is repeated until the corrections become negligible.

C. Interactive object detector/localizer and its calibration

The interactive sensor system is used for locating target objects for further handling or processing by robots. This is needed in such applications, where the target object geometry is well known, but environmental conditions or the size or

complicated shapes of the target objects does not allow the use of fully automatic sensor systems. The interactive sensor system consists of two cameras, attachable tags, and as such follows the principles of existing industrial 3D measurement systems. When the human operator attaches tags on 3D features, e.g., surfaces of the target objects, the camera system computes and records corresponding 3D points. When these are matched to a geometric reference model of the target object, the location of target object in the robot working space can be derived and used for on-line path updates [9].

Localizing the target objects using external fixed mount cameras is possible only if the positions and orientations between the robot and the cameras are known. The camera calibration means here determining the position and orientation parameters between the robot and the cameras. In general camera calibration includes also determination of internal camera parameters such as focal length and lens distortions; these internal calibrations are here assumed to be solved (for a solution, see e.g.[9]).

The cameras are located in fixed position in the robot cell which are calibrated to robot coordinate system with the help of a sharp ended calibration tool attached to the robot flange, in practice gripped by the robot. The sharp tool tip is easy to detect from images. Basically a free set of calibration motions are done and the corresponding pose values are stored as well as the locations of the calibration point in the image planes. In addition, we need a fair initial guess for the hand-eye parameters as well as the location of the calibration point in the robot hand. Fig. 3 illustrates this situation for one camera. The basic principle of the calibration algorithm is as follows.

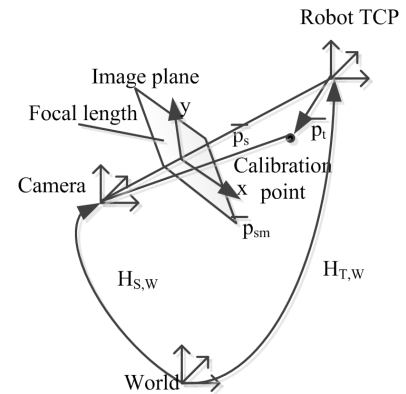


Fig. 3. The geometric relationships in camera / robot calibration. [4]

- Make a 'free set' of calibration motions and measure the corresponding points of calibration object end tip locations in the image planes.
- Fit the pose parameters of the cameras and the location parameters of the calibration object in the images so that the errors in the image locations are minimized.

The details of the calibration algorithm can be found from [9].

III. EXPERIMENTAL RESULTS

A. Calibration of the Eye-in-Hand sensor

In Fig. 4 arrangement for the hand eye calibration measurement is shown. The experimental setup includes KUKA Quantec series industrial robot of type KR120 R2500 pro with KR C4 controller (KUKA Robotics, Gersthofen, Germany) and MicroEpsilon LLT2700-50 2D laser scanner (Micro-Epsilon Messtechnik GmbH, Ortenburg, Germany). The laser scanner was mounted on the robot's wrist and the stationary calibration block was set on the floor. In order to carry out the hand/eye calibration the calibration block was scanned in five unique poses. The pose of the scanner with respect to the calibration block was estimated by fitting the CAD model of the block to the measured profiles including points from three intersecting planar facets. The respective robot mounting frame positions were acquired from the robot controller and the pose between the sensor and the robot mounting flange was estimated. In Fig. 5 examples of two scanning poses are shown. This use of highly precise regular object as target object represents the novelty of our work.

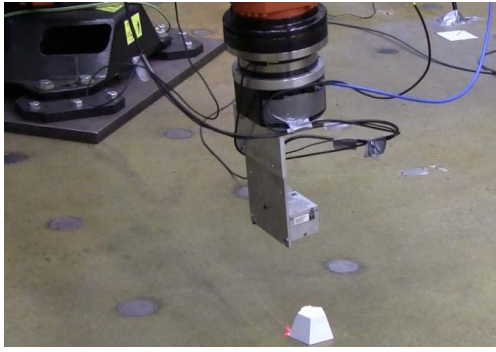


Fig. 4. The sensor unit in a test frame mounted in the robots wrist.

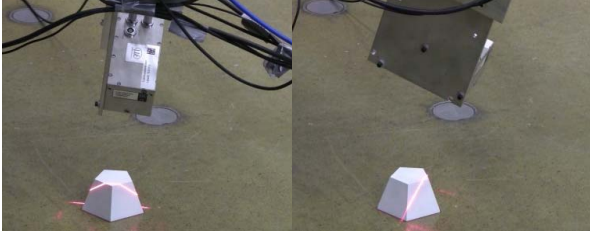


Fig. 5. Hand-eye calibration motions for the fine localization sensor.

Fig. 6 shows how the hand-eye parameters converge by the number of scans. The graphs on the right side represent the x, y and z parameters and rotation angles around x-, y- and z-axis. The measurement noise and the precision of robot positioning cause stochastic error which can be minimized by increasing the number of samples. This is clearly seen in the graphs as the variation of the parameters decrease as the number of samples increase.

The kinematic inaccuracy of the robot and manufacturing accuracy of the calibration block are most common sources for systematic error which cannot be evaluated from the graphs shown in Fig. 6. The systematic error can be evaluated by test measurements of known reference objects for example. Our

experience shows that with this calibration method at least ± 0.2 mm path accuracy for free (non-compliant) motions can be achieved in a limited robot work space if a sufficiently large sample group (>200 poses) is used. This is in line with the robot's repeatability accuracy and the accuracy of MicroEpsilon LLT2700 scanner. In this case 100 scans were used to gather enough data to calculate accurate enough tool correction for the scanner.

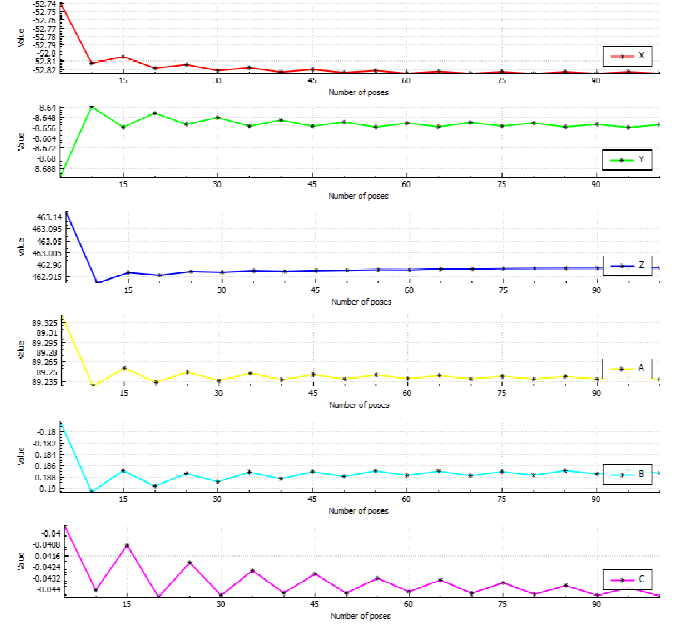


Fig. 6. Convergence of the hand-eye calibrations.

B. Calibration of the laser profiler in a fixed sensor station

Fig. 7 shows setup for measuring 3D points from a calibration plate's surface by moving the scanner along the rail and scanning at the same time. The calibration plane was measured in 10 different positions and orientations. Figure 8 shows the remaining errors after calibration. Clearly the effects of kinematic errors can be seen: at best there are sets of points where the residuals are between -0.5 and 0.5 mm (points from 20000 to 25000), but at worst between -4 mm and 1 mm (points from 9000 to 12000).

C. Calibration of the interactive multi-camera sensor system

The experimental setup included a robot with a pointer tool and two Canon D550 SLR cameras with Canon EF 24 mm f/1.4 lens. The cameras were mounted on the ceiling of the robot cell and the pointer tool was mounted on the robot's flange. The robot carried out a set of calibration motions, and the pose values and the locations of the calibration points (the tool tip) in the image planes were stored. Fig. 9 shows the calibration setup and the locations of the tool tip measured points (marked with O) and the model points (marked with X) after the calibration. Figure 10 shows the parameter convergence, which is good.



Fig. 7. Setup for calibration of the global laser scanner

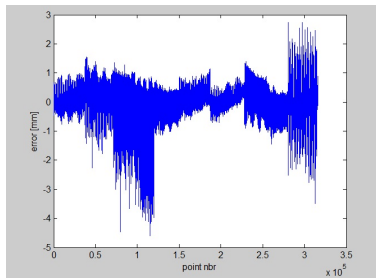


Fig. 8. Remaining error values (z-coordinate values in previous figures) after calibration for the 3D points used in calibration.

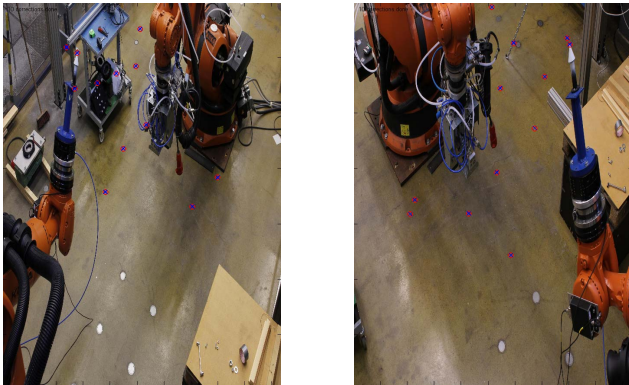


Fig. 9. Setup for calibration of the interactive multi-camera system

IV. CONCLUSIONS

We have presented calibration methods for object localizing sensor systems for robotic machining. The algorithms solve the coordinate transformations between sensor systems and the robot. The calibration procedures rely all on simple calibration objects and enable easy and flexible calibration. Practical tests showed that the performance with regards to accuracy was good and in expected levels.

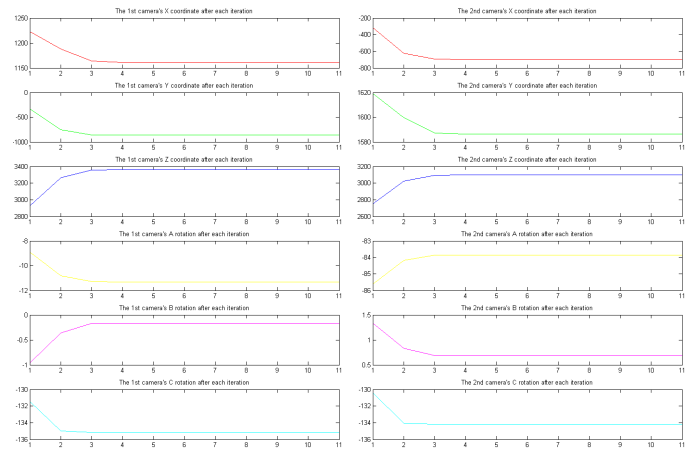


Fig. 10. Convergence of the residuals in cameras coordinates

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